

Technical paper

Adaptive edge finishing process on distorted features through robot-assisted computer vision

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ABSTRACT

Robotic deburring has emerged as a transformative solution in finishing processes, offering independence from operator influence to enhance product quality and reduce production costs. To do this, contour tracking is key to ensuring cutting tool contact and obtaining chamfers within narrow tolerances, especially on high added-value parts such as aeronautical turbomachinery. The persistent challenge involves accommodating workpiece variability, including deformations and burrs. Current probe positioning methods lack insights into burr morphology, while optical systems often require expensive scanners, resulting in time-intensive point cloud generation. Therefore, this paper presents an innovative industrial computer vision approach for adaptive deburring of functional edges. By processing the detected contour deviation signal, the proposed method generates tailored toolpaths based on real-time part conditions. The primary contribution lies in its ability to swiftly detect edges and burrs, dynamically adjusting machining toolpaths and conditions based on their dimensions, all through the rapid analysis of a simple image. This approach optimizes deburring operations, directing the cutting tool precisely where it is needed. Consequently, the default NC programming was enhanced with two new sub-programs tailored for deburring with conical tools—one for primary burr removal and another for uniform chamfering of distorted edges. The developed methodology can be easily applied in various industrial environments, offering a rapid and efficient means to improve chamfering of workpieces with variable contours. This research not only presents a practical solution for robotic deburring but also contributes to the advancement of adaptive manufacturing systems.

1. Introduction

“Remove burrs and break sharp edges” is a common note in all manufacturing drawings. Although currently edge-finishing process is automated in many industrial sectors, there are still many problems to solve for a correct implementation when dealing with components with difficult machinability, complex shapes and high added-value. This work is part of a broader research project that aims to advance in the automation of robotic edge-finishing processes for aeronautical components [1]. In precision deburring, when little material needs to be removed, contour tracking is of great importance to generate uniform chamfers. Despite this, conventional methods fail to cope with both the positional errors and the variability of burrs, which interfere with edge detection as well as the cutting process. In the vast majority of cases the burr is generated on edges formed by at least one flat surface. For this

reason, 2D-image processing techniques can be a fast, economical and easy-to-implement solution for both edge and burr recognition.

Nowadays, robots are used in a growing number of machining applications. A comprehensive review on the challenges and research aspects in robotic machining can be found in [2]. Among its advantages, highlights the improvement in process control, through automation, being more flexible compared to CNC machines, as well as providing a larger working area and lower operating costs. However, weak-stiffness and low positioning accuracy are still important factors restricting the rapid development of robots in the field of machining. Compared with some high stiffness of CNC machine tools, which is generally greater than 50 N/μm, the stiffness of industrial robots is less than 1 N/μm [3]. Low stiffness causes deformation and vibration in the machining process, resulting in poor machining accuracy and reduced surface quality [4]. Moreover, due to the existence of positioning errors, there is a

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certain error between the actual and theoretical machining positions [5].

Robotic machining can also be classified into low material removal rate (MRR) operations (including grinding, polishing and deburring) and high-MRR operations (involving milling and drilling) [6]. Accordingly, high-MRR operations are more stiffness-dependent, due to the higher forces developed in cutting. However, in the case of low-MRR operations such as deburring, positioning is key to achieving uniform chamfers across the part. This may well be the reason why robotic automation of deburring processes does not receive adequate attention, since it typically involves complicated tool motion trajectories for following workpiece edge contours while removing small amount of material.

Furthermore, when dealing with components manufactured using casting technologies; dimensional tolerances of component edges to be deburred are considerable. This means that accurate data of real edge geometry, as well as its position in space, is a key fact for the correct performance of the finishing process. This paper suggests an adaptive edge finishing process based on robot-assisted computer vision directly applicable to aero-engine components with distorted features. The main contributions and novelty of this work are summarized as follows:

- A vision-based technology is presented for distorted edge detection of aero-engine components, particularly those with unknown or distorted geometries.
- Using the workpiece contour signal obtained, smooth trajectories, and accurate burr size measurements can be performed.
- Programming an ad-hoc NC program to eliminate large localized burrs are included.
- These developments allow the generation of an adaptive chamfering NC program adjusted to the deformed contour.

This paper is structured as follows. In Section 2, a review of the related literature is presented, including works related with robot positioning, vision systems for manufacturing, application on turbo-machinery components and edge finishing processes. Section 3 introduces the experimental set-up and adaptive deburring methodology. In Section 4, developments for real edge detection using computer vision are presented. Section 5 gives more detail about burr identification and filtering. Section 6 includes the generation of new adaptive machining code for edge cutting with a defined-edge conical tool. Finally, the summaries are shown in Section 7.

2. Literature review

This section provides a literature review of the background analysed during the execution of this work. Specifically, it examines works on robot positioning in deburring processes, vision systems in manufacturing, and advancements related to breaking sharp edges and deburring.

2.1. Robot positioning in deburring processes

Since very small quantity of material is removed in deburring, minor positioning errors can lead to great discontinuities in the generated chamfers [7]. Differences between theoretical and true position values may be caused by both the robot and workpiece positioning.

To some extent, it is well known that robots present difficulties when it comes to following a given path. Not only due to the aforementioned lack of stiffness, but also because of other phenomena related to thermal expansion [8], hysteresis (backlash) [9], etc. These effects can be corrected or minimized by optimising the pose of the robot [10] or by using auxiliary cameras to track and correct its movements [11]. Thus, the application of a stereo camera system for external TCP-pose measurement may increase the absolute positioning accuracy of an industrial milling robot by up to 0.1 mm [12]. If a higher absolute accuracy of up to

0.03 mm is required, together with error correction along the tool path during feed motion, dynamically measuring laser tracker systems are the most appropriate solution. To quantify this, ISO standard 9283:1998 [13] describes the procedures to adequately measure robot position accuracy, repeatability and path accuracy.

Nonetheless, the main source of remaining errors is due to workpiece positioning. And not just because of its general location, but also because of the relative position of each point/feature with respect to its nominal value. Since each part shows unique variations in its dimensions, the robot must be able to locate and adapt to its morphology. There are multiple methods for programming industrial robots, usually divided into two categories: online programming and offline programming.

On the one hand, online programming is typically realized using a teaching programming method [14]. During teaching programming, the robot moves through the desired positions as the programmer enters in each step of an application through the teach pendant. Thus, this mode provides a natural approach to programming robotic systems, and does not require specific programming skills. Moreover, various novel online approaches were proposed to improve the efficiency of this method [15]. Although online programming provides users with a simple approach to generating a program for industrial robots, it involves taking the robot out of production, and a completely new program must be written if a new workpiece is processed. This is time-consuming and may lead to critical losses in production efficiency. On the other hand, offline programming is an automatic programming generation technique that does not require operations in real robotic systems, thereby greatly decreasing the downtime required for system programming, and resulting in enormous savings in labour costs [16]. Here, programs are created with the assistance of a computer using programming software and a simulated working environment. In this scheme, it is key to define the workpiece as realistically as possible. To do this, a CAD-based or vision-assisted approach can be adopted.

In CAD-based programming, user generates the robot path planning based on a CAD model of the workpiece [17]. Here, the CAD model with fixed geometry is placed in the simulated working environment by means of probe positioning. However, while a CAD-based offline approach can provide the complete geometrical features of workpieces to automatically generate a program, deviations between the workpieces and CAD model are not considered; therefore the accuracy in all the component features cannot be guaranteed. Some authors use CAD-based programming in combination with force control systems to adapt to the joint effect of burr variability and part distortion [7,18,19], even adapting the feed rate to maintain a constant contact force [20]. However, this requires a closed-loop operation with expensive equipment, which is also difficult to integrate and involves working with the same tool on edges with different burr conditions (which may exceed the maximum allowable material removal limit in areas with low burrs). Although there are developments that allow continuous inspection of the entire contour by sweeping free-form surfaces [21], they also fail to provide information about the size and location of the burrs, which greatly affects the deburring task. These drawbacks may be solved by employing vision-based offline programming.

2.2. Vision systems in manufacturing processes

Approaches found in bibliography use different types of vision sensors to capture the real geometrical features of workpieces in order to generate ad-hoc finishing programs [22–24]. Structured light scanners and 3D laser scanning are the most widely used technologies in this line. While laser scanning has its positive aspects, lasers are easily disrupted by reflective surfaces, any form of transparency, or even by complex interference patterns in the laser light. Normally, measurement accuracy of the point cloud is defined by the covered region of the structure light and the resolution of the CCD camera. In general terms, measurement accuracy is about 0.35 mm [23], although it is true that higher accuracies of tens of micrometres can be achieved using big-budget

equipment. Even so, structured light scanners are not free from reflections, even if they use short wavelength blue light.

When dealing with edges created by at least one flat surface, which are commonly found on turbomachinery components, it becomes possible to substitute costly equipment (such as 3D lasers and scanners) with a single camera, offering a more cost-effective and easily implementable solution in any industrial setting [25]. This approach eliminates the need for extensive point cloud data. Furthermore, image processing can be employed to identify distorted contours, burr size, and their locations. Vision cameras are already widely used in various industrial applications, such as tool wear monitoring, which utilizes low-resolution images with neural networks to determine if a tool is still suitable for use [26]. In measurement applications, in contrast, it is imperative to uphold a high image resolution, as the system accuracy will be determined by the number of pixels defining the region of interest and the quality with which the reference geometry has been obtained. Therefore, in this work CAD-based and vision-based offline programming approaches are integrated into a hybrid offline detection method [27], combining reference positioning by touch probing and visual detection in this case. The idea is to employ a contact probe to generate the workpiece reference system, to which results obtained by vision are then added. In this way, accuracy is maximised when obtaining the reference dimension for scaling. Although the application of vision systems to assist deburring operations is not new [28], the methodology proposed in this article not only aims to detect part contours, but also to adapt deburring strategy to the size of burrs in each section of the workpiece. In similar approaches, Wang et al. [29] employ the Dijkstra algorithm to re-position a robot according to points obtained by vision, based on a pre-planned path for automated local removal of weld slag defects. In the proposed work, this is done by pre-setting tools and their corresponding operating conditions for different burr sizes, to use them progressively only on the required working zones, in a more efficient way [30]. By this method, the aim is to improve part positioning while optimising cutting conditions in the deburring process, to achieve uniform chamfers across the entire surface.

2.3. Application case: burrs removal and break sharp edges

The driver of the proposed methodology and the target component of this work is the medium pressure turbine casing, as part of the next-generation aircraft engines [31]. This includes some embedded bodies, also known as bosses, distributed along the outer side. These present a variability in their morphology of up to ± 1 mm, typical deviations presented on cast components. A graphical representation is shown in Fig. 1. During manufacturing process, bosses are face milled and drilled for the coupling of the external motor connections, with this hole being the one used as a reference for the scaling in this work. Here, it is necessary to remove all the burrs generated in face milling, producing a

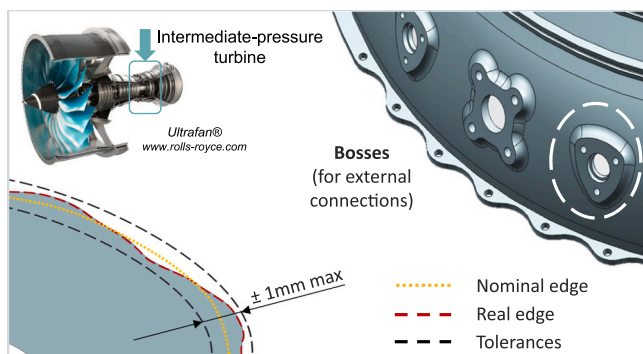


Fig. 1. Target case: bosses are cast components of the medium pressure turbine casing, which have deviations in position and shape of up to ± 1 mm.

uniform chamfer of between 0.1 and 0.4 mm, a task which, due to its difficulty, is currently performed manually.

Due to the problems associated with robotic machining, nowadays the most widely used strategy for the uniform deburring of difficult-to-cut and non-defined features is to split the task into 3 different steps [20].

First step is based on the removal of mayor burrs, as a preliminary step to chamfering. For this purpose, tools with high cutting capacity, such as cylindrical or barrel carbide tools, are used. Depending on the technology used, the first approach can be different. On the one hand, when dealing with the obsolete online programming approach, tools were manually adjusted to eliminate the main burrs taking special care not to damage other workpiece areas. On the other hand, with the rise of offline programming, these operations are now being performed assisted by force control devices to control the material removal. However, this requires working on the entire contour of the parts, even in areas with less burrs, and risks exceeding the maximum stock removal limit.

Second step consists of edge cutting, this is, chamfering or rounding, where the edge is given the required final shape. In this operation, contour tracking is more important. This is why in addition to the above-mentioned methods of force control compensation, other methods based on passive tool compliance can be used [32], just in order to achieve uniform results in spite of positioning errors of the robot.

Third step concerns an abrasive finishing process to smooth the secondary edges of the chamfer and to remove any smaller pieces of material that may remain on the part. Fig. 2 shows this sequence of operations applied to the type of parts presented in this work. Authors have a huge experience in this manufacturing field. In [33], flexible abrasive tools were tested and optimized to be a feasible and cost-effective alternative to eliminate turning pattern on brake disc surfaces, directly integrated into robotic systems. Furthermore, axial-compliant tools were shown to be able to control chamfer size on parts with burrs and robot inaccuracies [34].

As can be seen, each tool works in a different way in very specific contexts. Therefore, in order to improve results of compliant operations, either by force control or by passive tool compliance, this work proposes to focus the first edge preparation operation only on the workpiece areas with high presence of burrs. To this end, computer vision not only makes it possible to identify the deformed contour to improve the positioning of the tools during chamfering, but also to identify the main burrs and thus direct the machining paths only to these areas. This edge preparation allows the variability associated with burrs to be reduced, while the effect of deformations is also captured by the camera. Thus, tool compliance can be focused on correcting the robotic positioning errors mainly, improving the overall result of the process.

3. Experimental set-up and adaptive deburring methodology

The first step for the development of the technology proposed in this work was the design and manufacture of representative test parts. Real

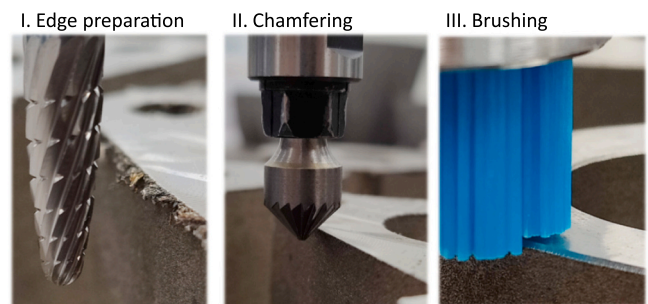


Fig. 2. Common 3-steps sequence for edge finishing: I) edge conditioning is performed removing larger burrs; II) chamfer shape is generated, and III) secondary edges are smoothed.

components are high value parts presenting confidential geometries. Thus, it is necessary to manufacture sample specimens that replicate conditions and characteristics of the real component, concretely the morphology of the bosses embedded along the engine casing. To simulate deviation caused by casting process, the outer contour of one of the parts was modified, creating an overstock material of 1 mm on one half of the part and the opposite on the other half. Both designs of parts were manufactured on Inconel® 718 (precipitation hardened, 45 HRC, ultimate stress of 1550 MPa), same material as used in real engine casing. The typical burrs of this kind of parts were generated by face milling under established cutting conditions (cutting speed of 700 m/min, feed per revolution of 0.10 mm/rev, and cutting depth of cut of 1.5 mm, with 4 RGN1207 SiAlON ceramic grade inserts mounted in a RNIW-03-06R toolholder. Finally, parts undergo a final hole making operation as part of the connections for the casing.

Image processing technologies have been widely applied in feature tracking based on imaging relationships [35]. In this case, computer vision is used to identify edge contours and determine their coordinates relative to the centre of the reference hole. In order to have an accurate measurement of the component edge, a contact measurement is taken as a reference and comparison with measurements obtained with the technology developed in this work. Fig. 3 shows this exercise when measured with a gantry structured coordinate measuring machine (CMM), with a measurement range of 900–1000–600 mm in the x-y-z axes respectively, and a resolution of 0.1 μ m. Continuous probing was performed using a 2.5 mm diameter ball. Results show that the real contour of the workpiece, besides being deformed according to the proposed design, is also displaced with respect to the hole centre. Thus, the part real deviation from the nominal geometry exceeds 2 mm at some points. However, it is important to note that, in order to perform this measurement, it was necessary to completely remove the burrs and use a CAD model with theoretical deviations included, to avoid hitting the probe stem. Following approaches developed in this work aim to address this limitation, in order to obtain similar results by using simpler equipment that also provides information about the state of the burrs.

Methodology developed in this work only requires the integration of an onboard camera, so that it can work in collaboration with the robot probe. Thus, the probe is in charge of generating the workpiece general coordinate system, which is used to locate the coordinates of the subsequently obtained image in the machine workspace. The rapid-measurement procedure performed by the probe is as follows:

1. First, a computer-aided manufacturing (CAM) project is generated based on the part nominal design (without deformations). This program then serves as a basis for the updated trajectories, maintaining the required robot pose configurations to enable these movements.

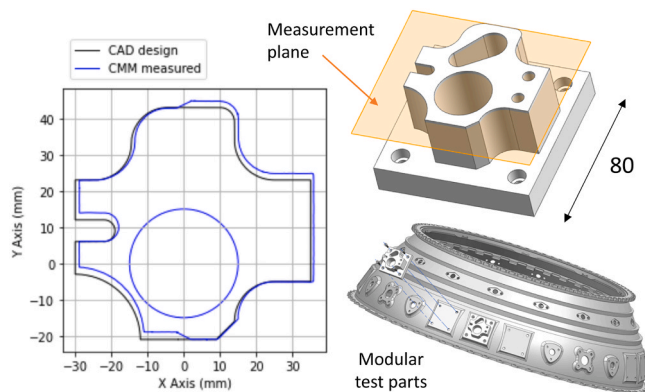


Fig. 3. Deformed workpiece contour measured by CMM, with coordinates referenced to the main hole. Sample parts present a modular design for easy interchangeability in the casing.

2. Using the measurement probe, the robot initially touches at least 3 spaced points on the face milled surface to locate the 2D image plane. This plane serves as a guide to update the robotic positioning and subsequently aligning the camera perpendicular to the part, in an orthographic projection.
3. Then, starting from the theoretical centre of the reference hole, the probe touches 8 different points around its perimeter; on the one hand, to locate the centre of the reference system of the piece and, on the other hand, to precisely calculate the diameter of the hole.
4. Finally, 2 points on a side face are touched to locate x-y axes of the reference system. To prevent contact with burrs, the shank is tilted at a certain angle.
5. A single image of the part is taken, processed, and then the original CAM program is modified according to these results, generating 2 new sub-programs for the roughing and chamfering operations.

Thanks to this approach, it is not necessary to measure the entire outer contour by probing, as this task is handled by the computer vision system. In this way, it is possible to obtain information regarding deviated contour and state of burrs. Fig. 4 shows a graphical scheme of the main operations required to generate new adaptive toolpaths following the approach outlined in this paper. Each of these operations is detailed in following sections of this document. All code necessary for image processing was implemented using an open-source programming language (Python). The main program developed imported a standard CAM script from a reference nominal part, and then generated 2 new adapted manufacturing programs with updated toolpaths: one only for burr removal and other for the chamfering of the deformed contour.

This approach was conceived for operations with conical cutters, separated at a certain distance from the edge, so cutting could be controlled by adjusting the axial position (z-axis). Thus, in roughing path, when not working on areas with large burrs, tooling was placed at a higher level, avoiding workpiece limits. This provided homogeneous conditions for the next finishing operation (chamfering), which was performed on more uniform conditions along the identified deviated contour.

4. Computer vision for real edge detection

This section outlines the steps taken to achieve accurate edge detection in distorted components. Identifying edges, even with burrs, is essential as a starting point for designing new adapted toolpaths. Therefore, the success of this process relies heavily on effective photo preparation and image processing. However, before starting with image capture, there are several considerations to be considered when aiming to obtain representative images of the real part geometry.

4.1. Camera lens distortion and transformation methods

Camera lens distortion, introduced by Conrady in 1919, involves the change in geometric representation of an object in the image plane [36]. This can pose challenges in machine vision applications, impacting the capture of part contour morphology. To address this, Brown introduced the Brown–Conrady model [37], categorizing lens distortion into radial and tangential types. The plumb line method was proposed for calibration. On the one hand, radial distortion is a general phenomenon caused by lens curvature, increases towards the image edges, causing straight lines in the real world to appear curved in the image. On the other hand, tangential distortion occurs when the captured object is at an angle to the lens, making the image appear tilted and stretched.

In this work, adopting a frontal arrangement during photograph capture simplifies the phenomenon, considering only radial distortion. This allows faster and easier calibration methods. Undistorted coordinates were calculated using known relative positions of a pattern (square corners in a 9 × 7 chessboard), both in the real world and in images captured at different positions, following Zhang's method [38].

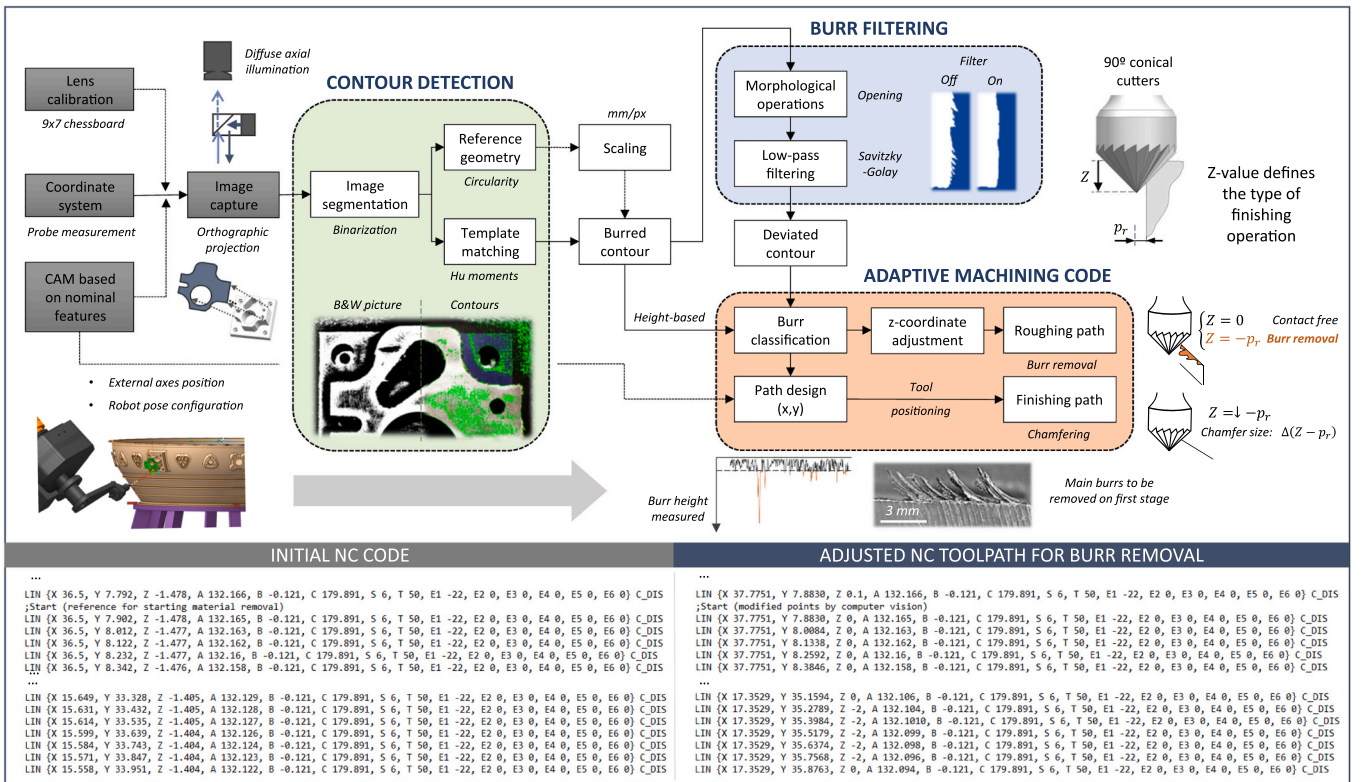


Fig. 4. Image processing scheme to adapt nominal-based toolpaths to deburring operations on real part. Adaptive methodology generates 2 new NC programmes: a roughing task for the main burr removal and a secondary accurate finishing stage for chamfer shaping.

For this calibration, a minimum of 10 different orientations is recommended. In other situations where the entire panel cannot be observed in all positions, ArUco calibration [39], based on fiducial markers, could be an alternative, though not applicable in this case.

4.2. Perspective distortion and correction techniques

Perspective distortion refers to the transformation of an object and its surrounding environment due to the proximity of that object to the lens and the observation direction. This causes changes in the apparent relative distances of the objects in the image, distorting their real dimensions and shapes. To minimise this effect, a forward viewing direction to the workpiece can be adopted at a sufficient spacing distance. With this approach, it could be considered as an orthographic projection model [40]. Thus, light beam is considered to travel parallel to optical axis. If the captured points lie in a parallel plane to the image level, the resulting photograph is essentially a scaled-down version of the original part.

Therefore, orthographic projection becomes a suitable approximation when the depth variation between the captured points is much smaller than the distance from these points to the image plane, i.e. the focal length. Thus, to minimise this distortion effect, camera positioning was controlled by orienting it frontal to workpiece, taking as a reference the surface generated after the face milling operation, at a separation distance of 500 mm in this case.

4.3. Light reflection and illumination

Since computer vision cameras catch the light reflected on objects, illumination is also key to enable correct image processing [41]. In the context of application of this work, the object to be highlighted is the machined surface of the part, so glare and reflections caused by this surface must be avoided. Therefore, diffuse axial illumination was used for this purpose. Thus, light was emitted laterally and reflected 90° by a

semi-transparent mirror that shifts light beams in the same direction as the camera axis, achieving a homogeneous diffuse light.

4.4. Image segmentation and binarization methods

After image acquisition, processing operations began to identify the workpiece outer contour and the reference feature. At this stage, the aim is to isolate the target surface from the background of the image, working on an initially unclassified multi-pixel matrix. To achieve this, certain deep learning methods can accurately measure image geometries when the object morphology is known (as demonstrated in [42], where yarn backbone contours align vertically). However, for general workpieces, these methods lack awareness of object attributes, resulting in segmentation predictions that deviate from expected characteristics, as seen with burrs [43]. In this context, pixel resolution in high-quality images determines both the coordinates of detected contour points and the height of burrs. Therefore, to address this, semi-automatic image segmentation is applied, by using edge detection algorithms tailored for industrial imaging conditions [44] to enhance processing efficiency and minimize the risk of overfitting [45,46].

Some of these popular edge detection algorithms rely on detecting changes in pixel intensity values, known as gradient operators. In grayscale images, each pixel is assigned an intensity value ranging from 0 (representing black) to 255 (white), based on the light captured by the camera. Among them, Sobel [47] method is simple and time-efficient, while Canny [48], although being more complicated, produces smoother edges due to implementation of non-maxima suppression and thresholding. However, when dealing with machined surfaces such as those in this work, surface patterns interfered with contour definition, leading to a splitting into different sub-contours. To obtain the full contour, binarization methods yield the most comprehensive results for these types of workpieces.

These methods are based on assigning each pixel to a certain group, by comparing its intensity with a given threshold. Thus, starting image is

binarized into two segments: image background and zones that exceed this threshold value. Then, an algorithm is used to find all visible contours in the binarized image, such as the one developed by Suzuki [49]. Therefore, choice of the appropriate threshold value is the key for succeed. If objects to be captured are easily distinguishable from the image background, and the image is properly illuminated, an overall threshold value can be assigned when performing the segmentation. To determine the most appropriate threshold value, there are methods providing automatic calculation through the histogram of the greyscale image, like the one proposed by Otsu [50]. This approach employs a weighted variance (σ_w^2), which is a measure of the overall variability or spread of pixel intensities within two segmented classes determined by a specific threshold value (t). The Otsu method is based on finding the threshold value (t) that minimises the weighted variance within the class given by the relation:

$$\sigma_w^2(t) = q_1(t)\sigma_1^2(t) + q_2(t)\sigma_2^2(t) \quad (1)$$

Where $q_1(t)$ and $q_2(t)$ are the probabilities of the two segmented classes (pixels with intensities below and above the threshold value), between 0 and 255. Thus, Eq. 1 is essentially a measure of how much the pixel intensities vary within each class, with the contributions of each class weighted by its probability. However, in cases where the image illumination gradient is high, or dark areas are presented (some examples will be shown in later sections), a different threshold value must be used for each area of the part by means of adaptive segmentation techniques [51]. To calculate local threshold value (t_L), either the arithmetic or the Gaussian mean of the pixel intensities within each region can be used. By the arithmetic mean, each pixel in the neighbourhood contributes equally to the calculation, whereas in the Gaussian mean, pixels further away from the centre of the neighbourhood area contribute less to calculation. The best results were obtained for the arithmetic mean:

$$t_L = \bar{I}_L - C \quad (2)$$

Where $\bar{I}_L$ was the mean of the pixel intensities within an image local neighbourhood area (a spatial region around each pixel in the image) and C a constant to tune the local threshold. Fig. 5 shows how this method allowed the full contour of the part to be highlighted against the image background, with no interference with machining marks. Thus, by applying the Suzuki algorithm, a complete contour surrounding the surface of interest (among all other contours that compose the binarized

image) was obtained. The size of the neighbourhood area for the formula was established empirically, in a preliminary sample image with the part set according to the operating conditions (illumination, surface characteristics, etc.).

4.5. Detected entities and target selection

During edge detection, not only those that were of interest (part outer contour and reference geometry) were shown, but also all the surrounding elements. In the case of Fig. 5(b), 98,381 entities were detected. Among all of them, based on the processing developed in this work, only the reference hole and the outer contour (including deformities and burrs) of the part need to be identified and selected. Thanks to having previously selected the segmentation parameters that allow obtaining a complete contour of the piece, it is not necessary to use complex matching algorithms subject to errors [52]. Instead, the geometric properties of each entity can be employed to calculate quickly and accurately the one that most closely resembles the target object, with only a single template required as a reference for each part.

4.5.1. Reference well-known feature for scaling

Reference geometry recognition, for the case of the main hole, was performed quickly after applying a series of conditions to all the detected contours. Fig. 6 shows the sequence of operations to be followed until finding the number of pixels that define the hole diameter. First, to reduce computing cost, all contours out of a reference pixel area range were ignored. Thus, potential contours were reduced up to 90%.

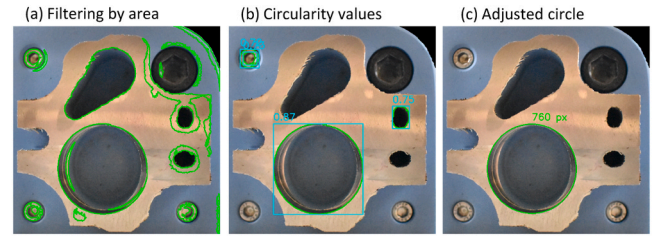


Fig. 6. Automatic detection process for reference hole identification: filtering by area (a), condition of minimum circularity (b) and measurement of the largest contour diameter (c).

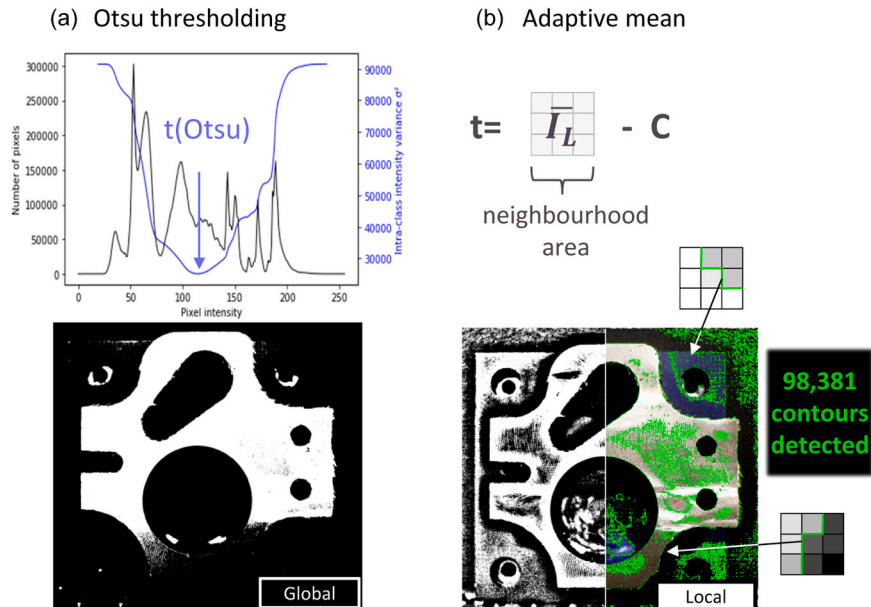


Fig. 5. Binarization methods: the Otsu method (a) does not allow full part contour to be shown, which is achieved by using adaptive mean binarization (b).

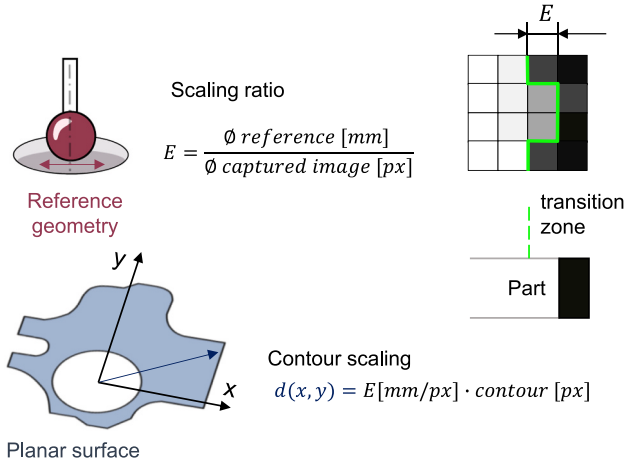


Fig. 7. Scaling the contour based on a part reference geometry. It conditions the accuracy of the boundary detection.

Then, a circularity condition C (Eq. 3) was applied to select those geometries with a result higher than 0.7 in this case.

$$C = \frac{Perimeter^2}{4\pi \cdot Area} \quad (3)$$

Finally, from all those detected that met the above conditions, the element with the largest area was selected and the diameter dimension in pixels of its circumscribed circle was calculated. As an example of such a case, for a hole of 29.984 mm diameter (previously measured using contact techniques), the 760 pixels captured corresponded to a scaling ratio of 39.45 $\mu\text{m}/\text{px}$.

A key aspect in this phase relates to image scaling, where the known physical characteristics of the part are correlated with the number of pixels captured to measure the remaining features of the image. In this case, the diameter of the hole measured with the contact probe is linked to the number of pixels in the circle calculated in Fig. 6 to obtain the value for each pixel (scaling ratio E).

In the gray areas of the contour (the transitions between the part and the background of the image), pixels have an intensity value very close to the segmentation threshold. This may result in some background pixels being considered part of the piece if they exceed this threshold. Since all pixels are associated with a scaling ratio E (mm/px), the smaller this scaling ratio, the less information is lost in edge detection, and more workpiece geometry can be identified.

Having obtained the physical dimension of the hole by probing, photo accuracy was maximised and the only remaining variable was the relationship between the camera resolution and the capture distance. For this work, a camera of 24 MP resolution (6 MP horizontal and 4 MP vertical) and a focal length of 18 mm was used, while shoot distance is explained later in this article.

4.5.2. Contour edge matching

In the case of the outer contour, the identification process is not so straightforward, as these boundaries do not conform to regular shapes. In this case, the procedure implies to employ a template with the nominal design of the part and select the contour edge that best fits it.

Among all the existing 2D object matching techniques [53] (such as the cross-correlation algorithm, sum of absolute differences, Hausdorff distance, etc.), the most complete and easy to compute is the one based on invariant moments [54]. Being immutable under linear coordinate transformations, invariant moments are useful features in pattern recognition problems. This algorithm is one of the most popular methods for image matching, as it allows different features of the part to be obtained regardless of rotation, scale and translation [55]. For two given images (that of part A and its reference template B), the following

matching is established:

$$I_1(A, B) = \sum_{i=1}^7 \left| \frac{1}{m_i^A} - \frac{1}{m_i^B} \right| \quad (4)$$

Being:

$$m_i^A = \text{sign}(h_i^A) \cdot \log|h_i^A| \quad (5)$$

$$m_i^B = \text{sign}(h_i^B) \cdot \log|h_i^B| \quad (6)$$

Where I_1 refers to the distance between two images (A and B). If the distance is small, the shapes are close in appearance and if the distance is large, the shapes are farther apart in appearance. Then, h_i^A and h_i^B are the Hu invariant moments of A and B respectively (a set of 7 moments that are invariant under translation, rotation, and scale transformations). These invariant moments are calculated from linear combinations of simple moments (previously normalised according to their scale). Being these simple moments the weighted average of the intensities of the contour pixels [56]. Moreover, in order to reduce computational cost, a condition of minimum area was previously established, achieving a reduction of up to 99% of the comparable base. Hence, potential contours of the part were related to that of the reference template, even when the former showed distortions due to deformations or burrs. Fig. 8 shows the different numerical results obtained for the template matching of manufactured test parts (b-c), being 1 for identical parts and 0 for contours with no common characteristics (the result of subtracting the distance I_1 between the images from the base geometry with a value of 1).

Fig. 9 depicts, among all the detected contours by the adaptive mean binarization, the one with the smallest distance I_1 to the nominal template in both deformed and non-deformed pieces. It is evident how the detected outer contours perfectly conform to the morphology of both pieces.

When analysing the boundaries of the detected parts, it becomes apparent that the highlighted outline encompasses the entire machined surface of both parts, including burrs. In Fig. 9(a), for the nominal part, a section of the burrs in the upper right zone remains unidentified due to being burnt portions, displaying a distinct colour tone from the rest of the surface. To ensure accurate experimental results, test parts were painted on their base to prevent the presence of oxides, a concern absent in real components where operations are conducted consecutively in controlled environments. Furthermore, in the real scenario, the contrast between the matte base of the cast casing and the shiny surface of the machining operations helps to highlight it against the image background.

5. Burr identification and filtering

Edge machining paths must avoid copying the noise-paths caused by burrs, so effective burr filtering must be performed while preserving the possible deviations of the casting. Most of the burrs present on the part have a serrated shape that can be removed by morphological operations. In addition, in order to smooth the resulting contour, in this work low-pass filters were introduced on the deviation signal between contour and nominal template.

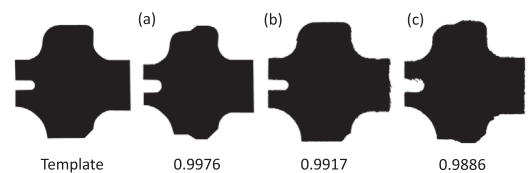


Fig. 8. Template matching: Result of comparing the template with (a) distorted design model, (b) nominal part with burrs and (c) distorted part with burrs.

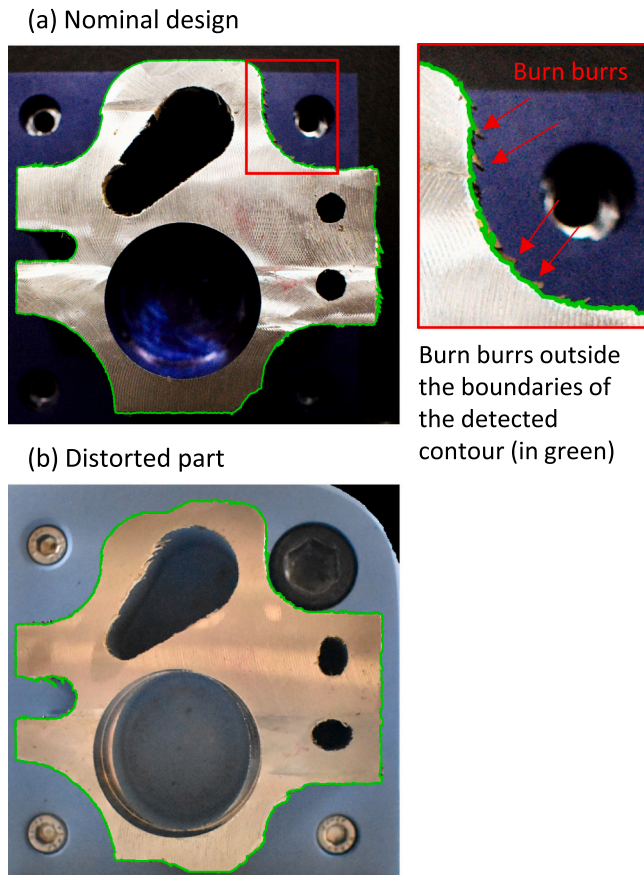


Fig. 9. Detection of burr contours on test pieces. From all the detected contours in Fig. 5, the one with the smallest distance I_1 to the nominal template is selected (based on the Hu invariant moments).

5.1. Morphological operations for burr detection

Morphological operations are those in which the shape is altered, by means of a structuring element, to produce a more defined form [57,58]. Most of these operations are defined from two basic processes: erosion and dilation.

Erosion removes the boundaries of the burr regions, reducing their area while maintaining the main shape. For this purpose, a structural element consisting of a uniform rectangular nuclear matrix kernel was used to cope with the random burr orientation in this work. By overlapping this kernel with the part boundaries, it was analysed whether all kernel pixels were completely contained within the workpiece image. In the gaps between the spike-like burrs, kernel pixels were not completely contained within these boundaries; therefore, the core pixel in the output image was set at zero, eroding the part edges. By selecting a kernel of sufficient dimensions, it was possible to completely remove the root of main burrs with these characteristics. To avoid eroding over the machining marks and creating false cracks in the part, the contour was filled beforehand to ensure that all the kernel pixels were contained within it. Finally, to restore the original dimension of the part (now free of main burrs), a dilation operation was applied at the boundaries of the contour. In this case, the process is similar, except that when at least one element of the kernel was inside the contour boundaries, the core pixel of the kernel in the output image became part of the piece. The concatenation of an erosion operation followed by a dilation is known as "opening", enabling the generation of burr-free contours as depicted in Fig. 10:

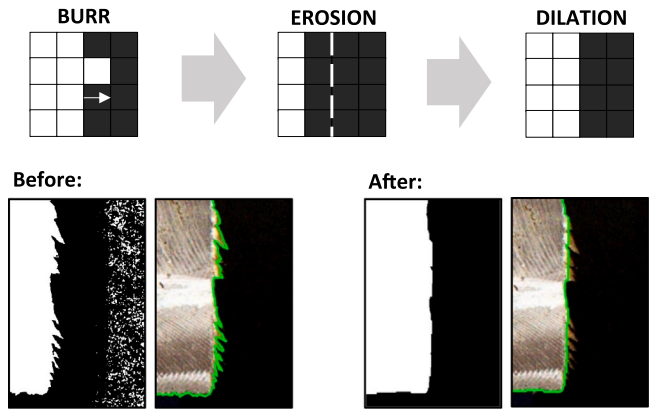


Fig. 10. Sequence of morphological operations: An opening is applied over a previously filled contour to avoid generating false cracks.

5.2. Smoothing of the contour deviation signal

Contour results obtained after morphological operations still present noise at the roots of the burrs that must be smoothed to avoid sudden movements during machining. As a novelty in this work, it was proposed to obtain the approximate values of contour deviation and then apply low-pass filters to smooth this signal, as if the "roughness" were eliminated.

Since machine vision allows a large number of contour points to be obtained (6503 points for a perimeter of 244.29 mm), deviation distance can be approximated as the minimum distance from each detected contour point to CAD points. To calculate these distances, a widespread algorithm is the ICP (Iterative Closest Point) [59–61]. This is commonly used to find the transformation that minimises the average distance between two sets of points, resulting in an approximate correspondence between them to align both clouds of points. However, this does not correspond to the minimum distance between each pair of points. For this reason, in order to improve result accuracy (by not requiring point transformations), the K-D tree for associative searching algorithm was used in this work [62]. As presented in Fig. 11, the tree was constructed based on the nominal design, recursively dividing the space into two halves at each level, using the mean value in one of the dimensions as a separator. Then, a nearest neighbour search was performed, in which the distance between each contour point and the tree points is compared to find the closest match (obtaining a contour deviation signal as depicted in Fig. 11).

Subsequently, low-pass filters were applied to allow the lower frequencies (part morphology) to pass through and attenuate the higher frequencies (burr noise). A basic example of such filters is the Butterworth filter [63]. However, for parts where the deformation gradient is high (as in the case of the deformed test part in this work), the Savitzky-Golay filter [64] proved to be more effective as it better adjusted to the relative maximum and minimum of the signal, as well as to the width of the peaks.

5.3. Procedure validation and comparison with other technologies

Once established and optimized the edge identification procedure, it is necessary to perform a validation exercise to verify the developed process by comparing results obtained with other contour identification methods. In this case, using CMM contact measurement as reference (which reveals measurement accuracy below $2\ \mu\text{m}$), edge identification process is compared with the computer vision technology proposed in this work. Therefore, results of these deviations give an idea of the computer vision accuracy compared to a more robust measurement method (CMM).

Additionally, test part was measured using a 3D scanning instrument

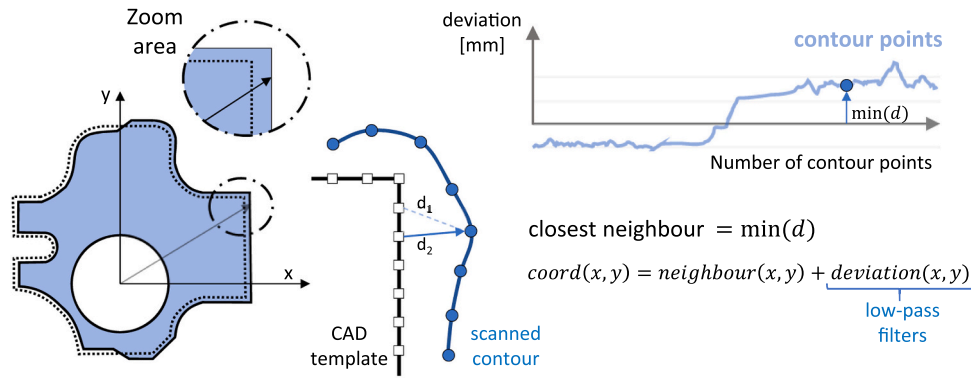


Fig. 11. Computing of the contour deviation signal from the minimum distance between the detected contour points and the CAD design.

to demonstrate the capabilities of industrial optical technology. The instrument was equipped with three class 2 laser beams with 660 nm wavelengths crossed at 120°, allowing it to capture up to 25,500 points per second and per beam. To achieve acceptable results, the surface was sprayed with anti-glare paint to reduce flaws in the generated mesh. For this technology, alignment was maintained as calculated in the probe operation and a section was made at a lower level similar to the one adopted for the probe path. Fig. 12 displays the results of both comparisons, using CMM data as the reference value.

Results demonstrate a significant improvement compared to the initial conditions shown in Fig. 3 when using the computer vision system. Initially, the part exhibited overall deviations of approximately ± 1.5 mm. Fig. 13 depicts the histogram of these deviations, initially

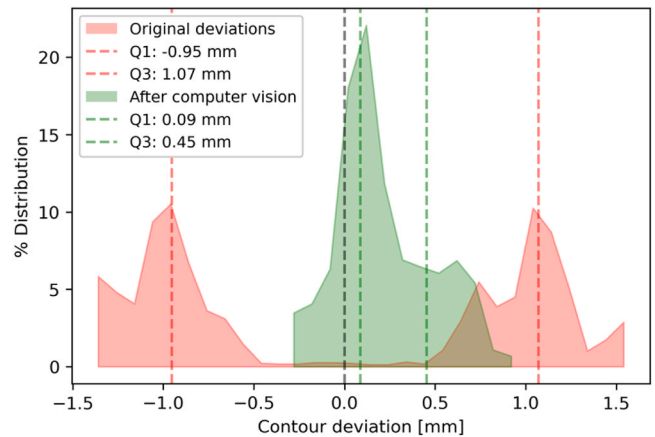


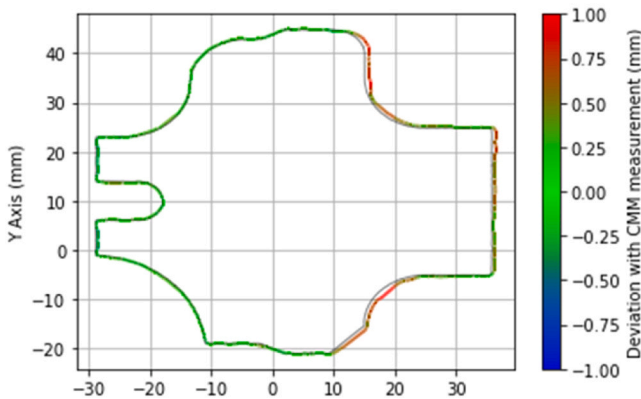
Fig. 13. Histogram of original deviations (red) and the computed deviations after image processing (green), for the deformed part. Auxiliary lines with the indicators of the first (Q1) and third (Q3) quartile are included for each case.

distant from the actual boundaries of the part, and demonstrates how these measurements converge towards the 0-value (black dashed line) using non-contact measurement methods. The root-mean-square error (RMSE) started in 1.050 mm for the original case, being reduced to 0.383 mm and 0.076 mm, for the artificial vision system and the scanner, respectively. Moreover, the median deviations of the detected contour points, compared with CMM measurements, are 0.194 mm and -0.021 mm, respectively. This indicates that the majority of contour points measured by the simple vision system successfully capture deviations of less than two-tenths of a millimetre.

The difference in the RMSE value between the artificial vision system and the laser scanner is due to the sensitivity of this indicator to extreme values. As observed in Fig. 13, the computer vision detects, in certain contour areas, points with a higher positive deviation. If these areas are analysed, it becomes evident that, although these boundaries do not correspond to the edges of the actual part, they are part of a uniformly distributed burr that had not been filtered during the image processing.

However, this is not a major concern for deburring, as these burrs still remain part of the piece. Instead of being removed in the initial roughing stage, they can be addressed during the chamfering operation that follows the same profile. For the remainder of the part, the contour deformation is accurately and uniformly detected in less time than the laser scanner (both in scanning and processing) and without preparatory steps such as anti-glare paint application. Neither the laser scanner nor the CMM probing recorded these uniform material build-ups, as they measured at a lower level, missing the opportunity to analyse any burrs. On the contrary, thanks to the information captured by artificial vision regarding the size of the serrated burrs, in the following section of this

(a) Computer vision



(b) Laser scanner

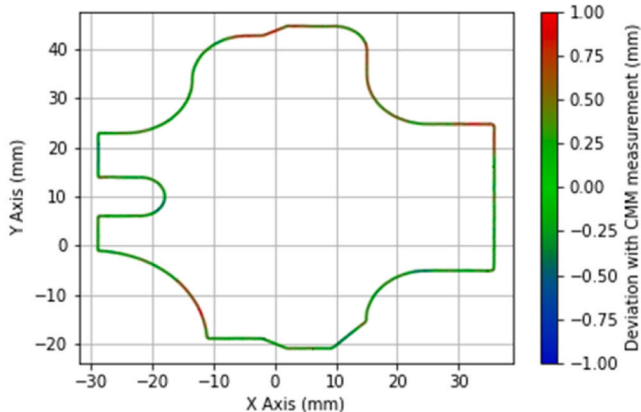


Fig. 12. Comparison with the reference detected edge (CMM) using: (a) computer vision and (b) laser scanning.

work, this data is used to design new specific trajectories to eliminate these burrs.

6. Generation of new adaptive machining code

The ultimate goal of this study is to generate new adaptive chamfering paths for edge cutting with a defined-edge conical tool. These paths are designed to accommodate both part deviations and the presence of burrs. To achieve this, the new chamfering trajectory corrects tool positioning, enhancing the uniform engagement between the tool and the workpiece. Before this operation, a roughing path is employed to adjust the tool depth of cut, effectively eliminating larger burrs and thereby enhancing the overall outcome of the subsequent deburring sequence.

6.1. Adaptive path based on deformed contour: design of the chamfering operation

Regarding the design of these new trajectories, coordinates of the detected contours cannot be used directly, since tool centre point (TCP) must be separated by a sufficient pre-set radial distance (p_r), being cutting speed at the contact point an appropriate value (at tooltip its value is zero). To perform this modification on the contour acquired by computer vision, and offset procedure was programmed. Concretely, code was based on projecting this distance in the normal direction to the line joining the neighbouring points to each original point of the contour. Thus, for each point i , the angle α of the line joining its previous point i_{-1} with its next point i_{+1} , in the forward direction, is as follows:

$$\alpha = \tan^{-1} \left(\frac{y_3 - y_1}{x_3 - x_1} \right) \quad (7)$$

The normal direction is applied towards the outside of the workpiece, being for an anticlockwise path:

$$\theta = \alpha - 90 \quad (8)$$

The offset radial distance p_r , set for operations with this tool geometry was set to 2 mm, to place the cutting radius in the middle of the available cutting edge, with the new 2D coordinates of the toolpaths being such that:

$$\begin{cases} x'_2 = x_2 + p_r \cos \theta \\ y'_2 = y_2 + p_r \sin \theta \end{cases} \quad (9)$$

The number of points obtained by machine vision (>6500 in this case), although allowed characterisation of the new part profile, were excessive to be transferred to robot movements. Therefore, instead of performing the operation show in Fig. 14 for each point of the contour,

this was first simplified to the minimum number of points required to define its boundaries (274 points after using the Douglas-Peucker algorithm for curve simplification [65]). Then an extrapolation to obtain a constant step of 0.1 mm (2640 points at the end) was performed, so the robot could chain movements with precision and agility. However, these new coordinates, although suitable for positioning the tool centre point (TCP) of the cutting tool in the workpiece reference plane, do not guarantee that the robot will perform the movements correctly. Among other things, this may be because anthropomorphic robots can reach a position by multiple axis configurations. To deal with this drawback, configurations of the closest points of the positions already verified from the previous nominal-based programming were used as a basis. In the case of the robot used in this work (KUKA KRC4), each configuration is determined from the S (Status) and T (Turn) values, which are calculated from a relationship between axes defined by the manufacturer. These values are kept fixed while deburring operation is performed, as the aim is to minimise movements between axes to maximise accuracy. Therefore, for the new deburring paths, S and T values were imported since changes introduced do not involve major robot additional movements. In the same way, values corresponding to tool tilt angles and rotation angle of the rotary table for casing positioning were also imported. Furthermore, air approaching and retracting movements were also maintained, avoiding collisions with other elements of the cell. In order to link the new detected contour to these movements, the point closest to the original starting point of the trajectory was found and set as reference for the new version. As this was slightly displaced with respect to the nominal position, the approach and retraction movements were also adjusted to coincide with the connection between the start and end points of the new trajectories.

Thus, these modifications made to original NC program make it possible to position the tool accurately with respect to the actual workpiece. Therefore, for precision chamfering with this tool geometry, it is only necessary to set a depth of cut (z) value higher than the pre-set radial distance (p_r), where the difference between these values will be the target chamfer size. These trajectories also serve as a basis for the design of the roughing operation, for which it is necessary to identify those zones with high presence of burrs.

6.2. Burr measurement and clustering

One of the great advantages of artificial vision applications involves their ability to detect defects or threats. Qin Hu developed a method to calculate ice thickness on wind turbine blades by computing height based on the pixel ratio with the image obtained from a calibration block [66]. Burr height in the context of the present work corresponds to the material dimension that departs from the part clean contour, which is similar to ice thickness measurement of the previous example. Its

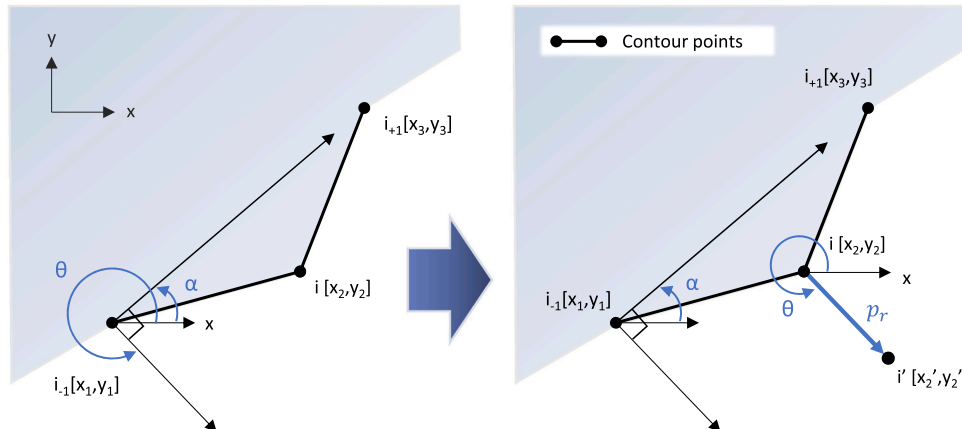


Fig. 14. TCP of the tool offset a distance p_r from the i point of the contour obtained by computer vision, normal to the line joining the neighbouring points (i_{-1}, i_{+1}).

measurement is straightforward, from the gap between the original contour and the one obtained after filtering. To obtain the corresponding values for the points of the new trajectory generated, the K-D tree search algorithm was trained with this set of points and then the minimum distance to the contour with burrs was measured, deducting afterwards the value of the side tool offset.

Fig. 15(a) shows burr height measured on the test piece with deviations. Regions with higher presence of burrs, such as the highlighted in red, hinder the machining of uniform chamfers, so they must be removed beforehand in roughing operation. To define which burrs need to be removed at this first stage, a minimum height criterion can be established. Fig. 15(b) shows the colour map with heights of the burrs detected on the part. Results show how spike-type burrs were correctly highlighted in red, while uniform material build-ups were hardly detected, being classified as deformations as shown before in Fig. 13. For these latter cases, burrs are removed directly in the chamfering operation.

A further useful approach to classification from the burr height plot is based on prominence. Prominence measures the vertical distance between a peak that stands out from the rest of the points in the burr height signal. According to this criterion, only burrs with characteristic points having a notable elevation or prominence of at least 0.5 millimetres with respect to their surroundings are considered significant. Fig. 15(c) illustrates the result of applying this criterion to burr detection, showing similar results to those shown in Fig. 15(b).

6.3. Adaptive path based on burr height: design of the new edge preparation operation

The approach for the design of the new deburring toolpath when dealing with critical burrs is based on an additional adjustment of the tool cutting edge. Concretely, the position of the conical cutting tool is

modified, passing separately from the detected contour of the part by default, and adapting its position in the z-coordinate in marked areas, removing all excess material outside workpiece boundaries.

With this tool geometry, if tool depth of cut (z value) is established equal to the pre-set radial distance (p_r), cutting edge is placed on the edge that matches the theoretical contour without burrs (after digital filtering), thereby reproducing this effect in a real manner by breaking the root of these burrs. The reason why this positioning is not taken along the entire contour is due to the fact that the error associated with the own robot positioning error is not considered in this work. Thus, risk is minimised by only acting on the zones with the greatest material build-up.

Therefore, the new trajectories for edge preparation and chamfering share the characteristic of adapting to the deformations of the workpiece, following the newly detected contour at a defined tool clearance distance. The difference between the two lies in the adjustment of the depth of cut, as the edge preparation operation also adapts to the size of the burrs on the workpiece. Fig. 16 illustrates the common CAM positioning (in the xy-plane) and the distinct depth-of-cut strategies based on the type of operation:

To sum up, on the one hand, the new toolpaths are adapted to the shape of the real part instead of the nominal design and, on the other hand, the most aggressive cutting procedure is applied only in the areas with larger burrs. This sets the stage for the subsequent chamfering operation, which follows the revised contour that is now free of major burrs, allowing for the generation of uniform chamfers. The computation time between image acquisition and the design of these new trajectories (with a 4-core processor at a working frequency of 2.70 GHz, and a RAM memory of 8 GB) was 31.7 s, less than the cycle time required for robot head change.

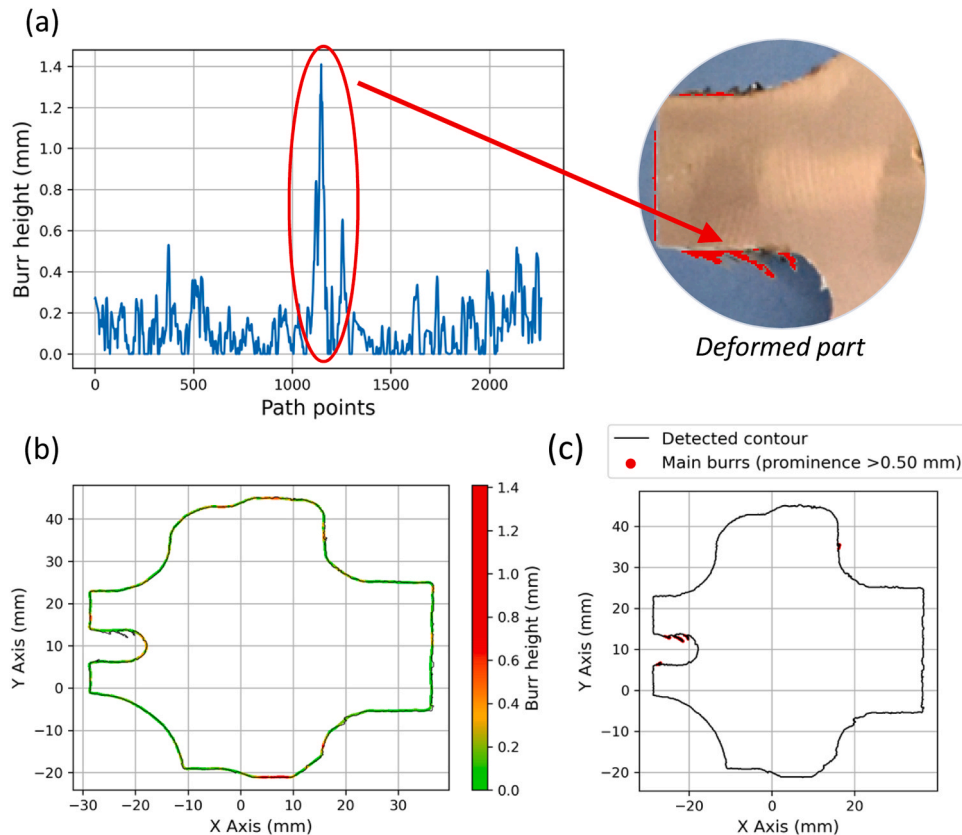


Fig. 15. (a) Measurement of burr height present on the deviated part, for each point of the new adapted NC toolpath generated at a separation p_r . (b) Colour map showing the location of largest burrs. (c) Detected burrs following a minimum prominence criterion (>0.5 mm).

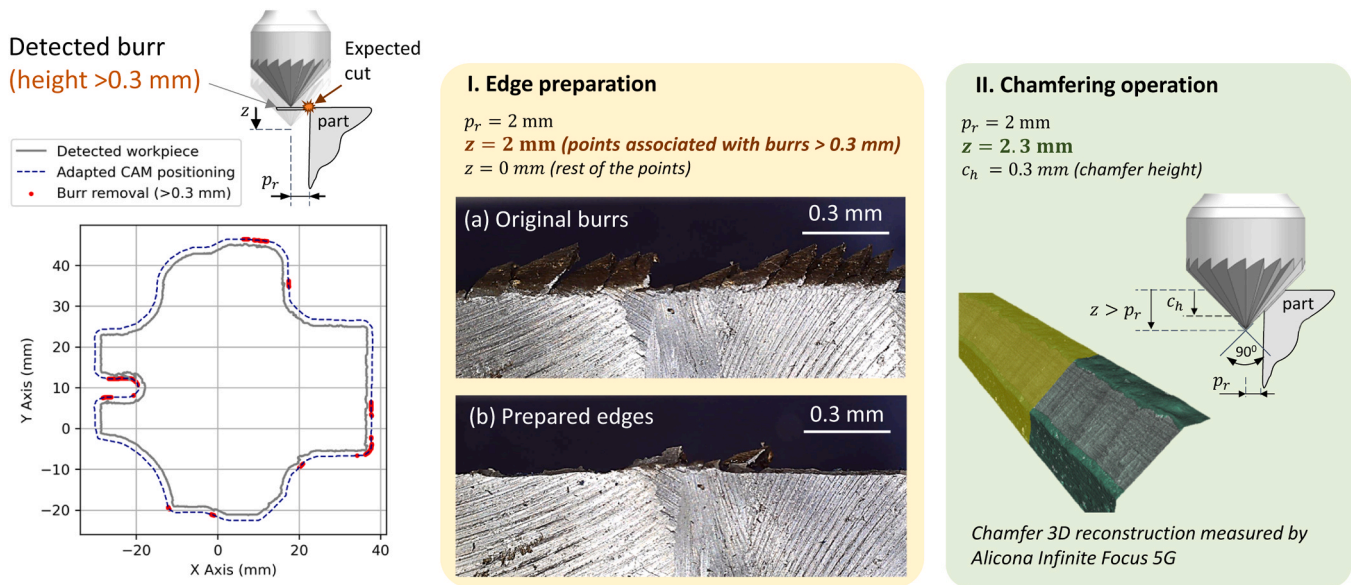


Fig. 16. Strategic depth adjustments for edge finishing: in the edge preparation (I), the depth of cut (z) is adjusted to cut only at the marked points (which are associated with the presence of large burrs), while in the chamfering (II), the depth of cut (z) exceeds the value of the pre-set radial distance (p_r), according to the desired chamfer size.

7. Conclusion

The objective of this study is to apply computer vision techniques to automatically modify a default manufacturing NC program, with two primary goals: i) addressing significant burrs in the initial stage, and ii) accurately following the actual edge contour in the subsequent processing stage. As a result, this work introduces a novel image processing methodology developed to create new deburring trajectories customized to real component conditions. This approach results in more consistent chamfers during machining on the detected edges, eliminating the presence of large burrs. Thus, burr detection was tested on two sample parts replicating the conditions of the actual target component, which includes bosses embedded in aero-engine casings. To validate the ability to adapt to geometrical changes, one of these test parts was also manufactured with deviations of up to 2 mm from its original CAD design, to compare the captured outer contour with the results obtained using a CMM machine. Based on the results obtained, the following conclusions are drawn:

- The developed computer vision system in this project benefits from the references previously obtained by a measurement probe. This hybrid approach only requires a few points to be measured with the probe, just enough to accurately locate the reference system and measure the hole dimension, while the features of the deformed contour and burrs are captured by the vision system, bringing together the best of both technologies. Moreover, the xy-plane touched by the probe serves as a guide to align the camera in an orthographic projection perpendicular to that plane.
- In contour detection, to ensure that detailed information on the contour of the workpiece is captured, a semi-automatic segmentation is proposed. Thus, the parameter settings for the detection algorithm must be tailored to both machine and workpiece lighting conditions. For this purpose, a preliminary capture is performed to adjust parameters and then repeated for the rest of the part batch. For the present conditions of this work, the adaptive binarization algorithms for local segmentation gave the best results, due to their ease in separating contour boundaries from surface machining marks.
- This adaptive machining approach uses a workpiece template that integrates the CAD design image and a basic NC program in the same coordinate system as the probe-generated one. This template

identifies the outer contour by comparing Hu invariant moments with all entities in the workpiece image, selecting the most similar contour. It is also used to obtain the contour deviation signal, subsequently processed with filters to smooth toolpaths during machining. The input NC code serves as a reference for the robotic macro-movements and tool orientation angles.

- Capturing the part with a 24MP front-facing camera at a 500 mm distance results in a scale of $39.45 \mu\text{m}/\text{px}$, effectively minimizing errors caused by pixels in the grey transition zone of edge detection. Through image processing, the initial deviations of $\pm 1.5 \text{ mm}$ are reduced, converging towards the actual boundaries of the workpiece. The root-mean-square error (RMSE) decreases from the initial 1.050 mm to 0.383 mm , with most points concentrated below two-tenths of a millimetre. These outcomes closely resemble those achievable with a comparative laser scanner, under ideal working conditions with the piece free of burrs and painted with anti-glare spray. The advantages of image processing, aside from not requiring any prior conditioning, include the automatic filtering of most burrs while processing data faster and at significantly lower costs, as pixels are utilized instead of point clouds.
- Sharp burrs with a saw or spike-like appearance are effectively filtered by morphological operations for the adapted CAM positioning. This allows for surgical removal during the edge preparation operation, targeting only the largest ones based on a criteria of minimum burr height or prominence. However, uniform material build-ups are more complex to filter and are subsequently processed as part of the workpiece deformed contour. For these cases, the new chamfering trajectory on these boundaries gradually removes material up to generate a uniform chamfer within the required dimensions.

Throughout this work, the necessary image processing techniques have been applied to address this current challenge, laying the groundwork for achieving full automation of the deburring processes for components of aeronautical engine turbines. If the segmentation provided by deep neural networks can be improved to accurately highlight the fine burrs details, they could be introduced as the primary segmentation method for such tasks as well. The authors expect that this work might serve as a foundation for future developments in this area.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mikel Gonzalez reports financial support was provided by Spain Ministry of Science and Innovation. Mikel Gonzalez reports financial support was provided by Basque Government. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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